SOUND PLACEMENT

User Guide · Version 0.1 · VST3 / Standalone

Welcome

Sound Placement puts a sound where you want it in a virtual room. Instead of juggling a panner, an EQ, a delay and a reverb to create a sense of position, you drag a single dot: left/right sets the stereo panning, and up/down moves the source closer to or further from the listener. All the acoustic cues that make your ear believe the placement are handled for you.

It runs as a VST3 effect in any compatible DAW (FL Studio, Reaper, Cubase, Ableton Live 12+, Studio One and others) and as a standalone application for quick experiments. Mono and stereo inputs are supported; the output is always stereo.

Installation

Copy `Sound Placement.vst3` to your system VST3 folder and rescan plugins in your DAW:

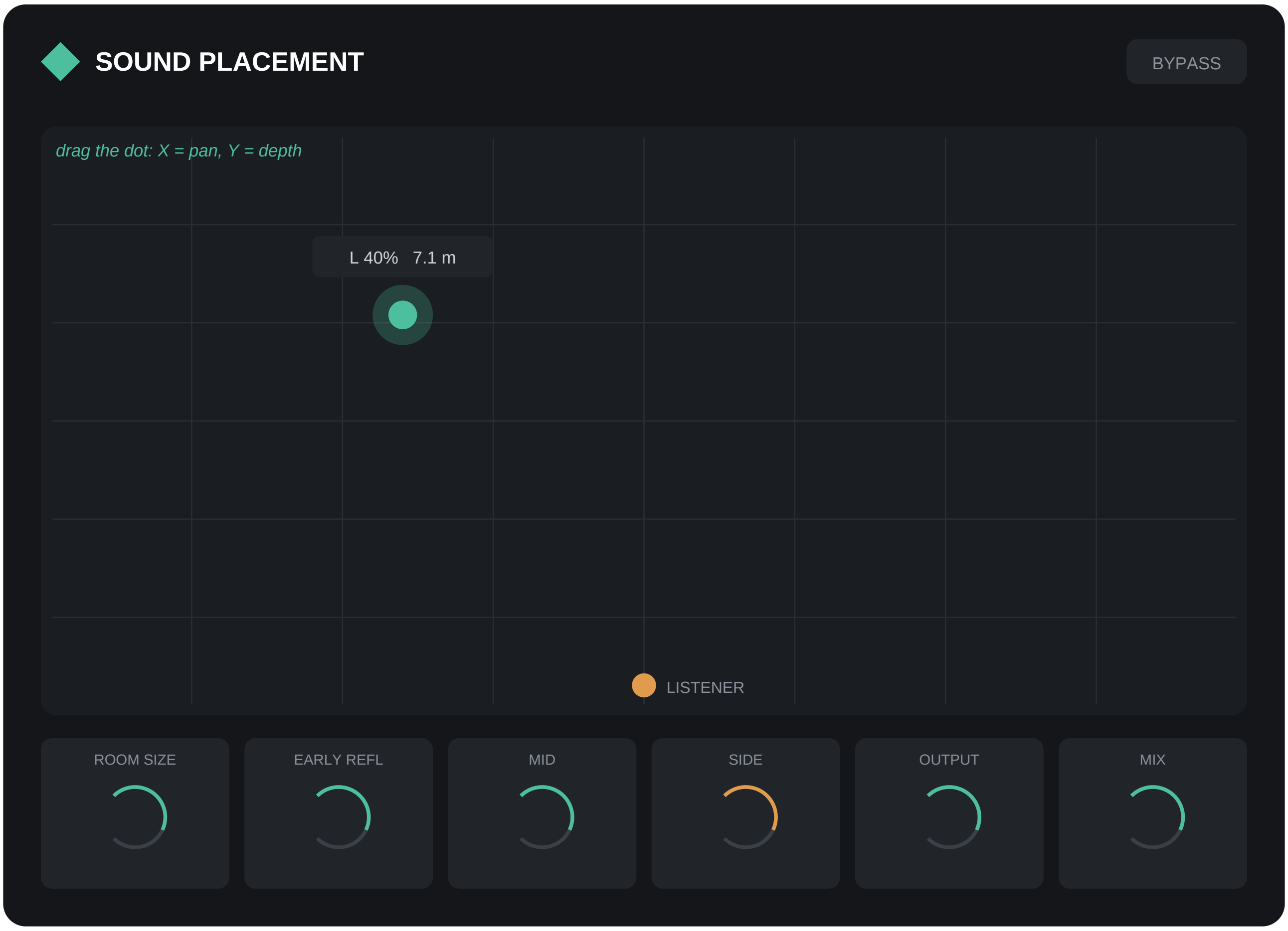
Windows	C:\Program Files\Common Files\VST3
macOS	/Library/Audio/Plug-Ins/VST3 (or ~/Library/… for one user)
Linux	~/.vst3

If you build from source, the build step installs the plugin into the correct folder automatically. See the README in the repository for build instructions.

Quick start

Insert Sound Placement on a track, play some audio, and drag the green dot around the room. That is genuinely all there is to it - the six knobs below the room are refinements, not requirements.

The room view



The large panel is the room seen from above. The orange dot at the bottom edge is the listener - you. The green dot is the sound source. Drag it anywhere: the horizontal position sets panning, the vertical position sets depth. Hovering over the room shows a readout with the current pan amount and the straight-line distance from listener to source in metres (the room is modelled as 10 × 10 m).

Both position axes are automatable parameters (*Pan* and *Distance*) in your DAW, so you can draw movement through the room over time.

What happens as the source moves back

Your ear judges distance from several cues at once, and the Distance axis drives six of them simultaneously:

Level	The source gets up to 15 dB quieter at the back of the room.
Air absorption	High frequencies fade with distance - a lowpass filter falls from 20 kHz down to 1.8 kHz.
Stereo width	Distant sources sound more point-like, so the stereo image narrows with depth.
Early reflections	The first bounces off the walls grow louder relative to the direct sound.
Reverb amount	The room's reverb tail rises with distance - the strongest single cue.
Predelay	Near sources leave a ~30 ms gap before the room answers; far sources almost none.

Parameter reference

Parameter	Range	Default	What it does
Pan	-1 … 1	0	Left/right position. Constant-power pan law, so the centre never dips.
Distance	0 … 1	0	Depth in the room. Drives level, tone, width, reflections and reverb together.
Room Size	0 … 1	0.5	Reverb decay time (RT60 0.4-3.0 s) and early reflection spacing.
Early Reflections	0 … 1	0.4	Level of the first wall reflections. Scales with Distance.
Mid Level	-60 … +6 dB	0 dB	Gain of the mid (L+R) signal - the centre of the stereo image.
Side Level	-60 … +6 dB	0 dB	Gain of the side (L-R) signal - the stereo edges. -60 dB = mono.
Output	-24 … +12 dB	0 dB	Output trim, applied after all processing.
Mix	0 … 1	1.0	Global dry/wet. 1.0 = fully processed, 0 = untouched input.
Bypass	on / off	off	True bypass. Also linked to your DAW's own bypass control.

All parameters can be automated from the host. Double-click a value box to type an exact value.

About the reverb

The room sound is a convolution reverb running a synthetic impulse response: exponentially decaying noise that is progressively darkened through the tail, mimicking how air and walls absorb high frequencies. Turning Room Size rebuilds the impulse response in the background without clicks or dropouts, so it is safe to adjust during playback.

Tips

- **Depth for a mix, not just an effect:** even a small Distance value (0.1-0.2) on a few tracks creates front-to-back layering that flat mixes lack. Save the deep settings for special moments.
- **Backing vocals behind the lead:** keep the lead voice near the front, push the backing vocals to 0.3-0.5 - they will sit behind the lead without any EQ carving.
- **Narrow it yourself:** Side Level at -60 dB gives you mono - useful for checking mono compatibility without leaving the plugin.
- **Automate a walk-past:** automate Pan from left to right while Distance dips and returns - the source appears to walk past the listener.
- **Parallel depth:** lower Mix to ~0.5 to blend the placed sound with the dry original - subtle depth without losing presence.
- **Gain staging:** heavy Mid/Side changes or a big room can change loudness; use Output to bring the level back in line.

Troubleshooting

The plugin does not appear in my DAW.

Make sure the .vst3 file (or bundle folder) is in the system VST3 directory listed above, then rescan plugins. In some DAWs the plugin list must be refreshed manually in the settings.

The sound seems quiet.

That is the point! Distance attenuates up to 15 dB. Pull the dot forward or raise Output if the placed level is too low.

Everything sounds distant/washed out.

Check that Mix is where you expect it and that Distance is not automated to a high value from an earlier experiment.

No stereo movement on a mono track.

The plugin outputs stereo. If your DAW forces a mono insert chain, place the plugin on a stereo bus instead.
